- Compute Match — Question and Answer Bank
  - Admin — alerts
  - Admin — analytics
  - Admin — custom-request
  - Admin — global-capacity
  - Admin — onboarding
  - Admin — review
  - Admin — timers
  - Any — feedback
  - Any — help
  - Any — lifecycle
  - Any — overview
  - Any — roles
  - Any — sign-in
  - Any — support
  - Any — timers
  - Any — whats-new
  - Customer — custom-request
  - Customer — request
  - Customer — search
  - Customer — tracking
  - Developer — api
  - Developer — architecture
  - Developer — delivery
  - Developer — git
  - Developer — infrastructure
  - Developer — keycloak
  - Developer — local
  - Developer — observability
  - Developer — salesforce
  - Developer — testing
  - Developer — workflows
  - Ncp — dashboard
  - Ncp — magic-link
  - Ncp — manage
  - Ncp — roles

- Ncp — sign-in
- Ncp — upload

Compute Match — Question and Answer Bank

Admin — alerts

Q: Is there a daily alert for requests near their deadline?

A: Yes. A Temporal schedule (in-process-requests-alert-daily) emails Platform Admins at 09:00 UTC every day with In Process offerings approaching their contract deadline.

Admin — analytics

Q: What analytics dashboards exist?

A: Two Superset dashboards: Compute MatchMake System Metrics (request, user and capacity analytics such as conversion rate, time to contract, capacity by platform and region) and Feedback (ratings, reasons, journeys).

Q: Who can see the Superset dashboards?

A: Users with the Keycloak role superset_dashboard_viewers (included for Platform Admins), mapped to the Superset DashboardViewer role, which can view the analytics schema only with no SQL Lab.

Admin — custom-request

Q: How do I process a custom request as an admin?

A: Open Custom requests, click the row, move it to In Review, then Mark as Matched (optionally recording the NCP, region and comment) or Close as Unserviceable. The customer is emailed each step.

Admin — global-capacity

Q: What is the Global Capacity Explorer?

A: A read-only map, list and chart of GPU capacity across all NCPs. Filter by Platform (needed for the chart), region, NCP, country and dates; export to Excel or CSV. Requires global_capacity_view.

Admin — onboarding

Q: How do I onboard a new NCP?

A: Go to Onboard NCPs, download the onboarding CSV template, fill in organizations and users, and upload it. A background workflow creates Keycloak organizations, users, roles and group memberships and sends welcome emails. Track progress in the history table.

Q: How do I approve a partner access request?

A: In Onboard NCPs, open Review Onboarding Requests, then Approve (provisions the organization and user) or Reject with a reason (emailed to the applicant). Requires manage_ncp and the NPN integration enabled.

Q: What are the NCP onboarding job statuses?

A: QUEUED, VALIDATING, VALIDATION_FAILED, ONBOARDING, COMPLETED and FAILED.

Admin — review

Q: How do I approve future capacity?

A: Open Review Future Capacities, go to the Pending tab, click Accept on the row and Confirm in ‘Approve Future Capacity Submission’. The capacity goes live for customers.

Q: How do I reject future capacity?

A: In Review Future Capacities, click Reject on a Pending row, enter a reason (required, up to 500 characters) and click Submit Rejection. The NCP is emailed the reason.

Q: Are future capacity submissions ever auto-approved?

A: No. They wait until a Platform Admin reviews them.

Admin — timers

Q: How do I extend a request deadline?

A: In Reservation requests (table view), click the row's ⋮ menu, choose Extend Acceptance Duration or Extend Contract Negotiation Duration, enter 1-200 days, check the previewed date and click Confirm. Needs accept_contract_extension.

Q: Can an expired offering be revived?

A: Yes. Extending the acceptance or contract deadline can move an Expired offering back to its previous active status.

Any — feedback

Q: How do I give feedback?

A: Click Share Feedback in the header, or answer the star-rating prompt after creating a request or uploading capacity. When cancelling you may be asked for a reason.

Any — help

Q: Where is the user manual?

A: Click the book icon (User Manual) in the header. It opens the full manual filtered to your role. The ? icon shows help for the current page.

Any — lifecycle

Q: What are all the reservation request statuses?

A: Open, Awaiting NCP Acceptance (AwaitingProviderAcceptance), In Process, Contracted, Rejected, Expired, Cancelled, plus In Review, Matched and Unserviceable for custom requests. The UI also shows Awaiting Provider Approval and Awaiting Customer Approval for counteroffer flows.

Q: What does In Process mean?

A: At least one NCP has accepted the request and negotiation is under way. The contract must be confirmed within 45 days by default.

Q: When does a request become Expired?

A: When NCPs do not accept within the 7-day acceptance window or do not contract within the 45-day contract window. When every offering has expired, the request is Expired and the customer gets one consolidated email.

Q: What happens to other NCPs when one NCP contracts?

A: Their offerings on the same request are Cancelled and they receive a ‘superseded’ email.

Q: When is a whole request Rejected?

A: When every NCP offering on the request is rejected.

Q: What is the difference between reservation status, offering status and GPU offering status?

A: Reservation status covers the whole customer request. Offering status covers one NCP’s slice of the request (AwaitingProviderAcceptance, InProcess, Contracted, Expired, Rejected, Cancelled). GPU offering status is the inventory listing status (ACTIVE, PENDING_REVIEW, REJECTED).

Q: What is a proposal?

A: A version of the reservation terms: dates, GPU count, platform, region and price. The customer’s request is the first proposal; NCP counteroffers add more. Price-only changes update in place; material changes create a new proposal in history.

Q: What is an allocation?

A: The capacity consumed when a request is contracted. Its statuses are scheduled, provisioning, running, stopping and complete.

Any — overview

Q: What is Compute Match?

A: Compute Match (the Compute MatchMake System) is an NVIDIA platform that connects customers who need GPU compute capacity with NVIDIA Cloud Providers (NCPs) who operate GPU infrastructure. Customers search clusters and send reservation requests; NCPs accept, negotiate and contract them; NVIDIA Platform Admins run the marketplace.

Q: What does NCP stand for?

A: NCP means NVIDIA Cloud Provider, also called GPU Compute Provider: an organization that owns and operates GPU data centers and publishes capacity on Compute Match.

Q: What are the user personas in Compute Match?

A: There are three: Customer (also called Offtaker), NCP (provider), and Platform Admin (NVIDIA staff). A user with both customer and NCP access can switch with ‘Switch profile to’ in the account menu.

Q: What is NCPMS or Clustruffle?

A: NCPMS is the Jira project key and internal name for Compute Match. Clustruffle is an internal code name that appears in metric names and Keycloak endpoints.

Q: Explain the end-to-end workflow of Compute Match in simple steps.

A: 1) NCPs upload GPU capacity via CSV. 2) Customers filter clusters on Find a Cluster to Reserve. 3) Customers submit a reservation request to one or up to 3 NCPs, or a custom request. 4) Each NCP gets a magic-link email and accepts, rejects or counteroffers. 5) The first NCP to click Confirm Contract wins; the request becomes Contracted, others are cancelled as superseded, capacity is allocated and the deal syncs to Salesforce.

Any — roles

Q: How do I switch between customer and NCP profiles?

A: Open the account menu (your name, top-right) and use ‘Switch profile to’. Menus and help update to the active profile.

Q: A menu item is missing for me. Why?

A: Menu items only appear if you hold the matching scope. For example Upload Capacities needs manage_inventory and Global Capacity Explorer needs global_capacity_view. Ask your Platform Admin to grant the role.

Q: Who can belong to more than one organization?

A: Only account-manager roles (account_manager_consumer and account_manager_provider). This is enforced by the multi-org-policy Keycloak extension.

Any — sign-in

Q: How do I sign up or log in?

A: Click Sign up or Login on the home page, enter your work email on the login page and click Login. You are redirected to Starfleet (NVIDIA SSO) and your company's identity provider, then back to Compute Match. Sign up and login use the same flow.

Q: Can I use my Gmail account to sign in?

A: No. Personal email domains such as Gmail, Yahoo and Hotmail are blocked. Use your work or enterprise email.

Q: What is Starfleet?

A: Starfleet is NVIDIA's single sign-on service. Compute Match uses it through Keycloak to authenticate users with their work identity.

Q: Why am I seeing Access denied?

A: Your identity was verified but your account is not authorized under current access policies (for example missing role or disallowed domain). The page shows your email and the reason. Contact computematchmake@nvidia.com, or sign out and use a work account that has access.

Q: What does 'Thanks for Signing up' mean?

A: Your account status is PENDING and awaits activation. You will receive an email when it is activated.

Q: What fields are needed to complete my profile?

A: First Name, Last Name, Organization Name and Phone Number (digits only, +1 pre-filled). Click Submit.

Q: How do I accept the terms and conditions?

A: A blocking modal appears after login. Scroll to the end, tick 'I have read and agree to the Terms and Conditions' and click Accept & Continue. You can also choose Sign out.

Q: Where do I land after logging in?

A: Customers land on Find a Cluster to Reserve, NCPs on the Dashboard (/provider/dashboard), and Platform Admins on the Global Capacity Explorer when available.

Any — support

Q: Who do I contact for access problems?

A: Email computematchmake@nvidia.com, or ask your Platform Admin to grant the role you need.

Any — timers

Q: How long does an NCP have to accept a request?

A: 7 days by default (MULTI_PROVIDER_ACCEPTANCE_EXPIRY_DAYS). A reminder is sent 3 days before the deadline.

Q: How long is the contract negotiation window?

A: 45 days from acceptance by default (MULTI_PROVIDER_IN_PROCESS_EXPIRY_DAYS), with a reminder 3 days before.

Any — whats-new

Q: What is the What's New banner?

A: A release highlight banner shown on your landing page for about 7 days after a release, tailored to your persona. Dismissing it keeps it available under the bell icon (or What's New in the menu). Use Mark All as Read to clear the unread dot.

Q: What is new in the October 2026 release?

A: Customers: faster request submission and real-time custom request tracking. NCPs: clearer upload status and sign-in from the NPN portal. Admins: the full custom request workflow (Open, In Review, Matched or Unserviceable).

Customer — custom-request

Q: What is a custom request?

A: A reservation request with no specific cluster selected. NVIDIA Platform Admins broker it manually, moving it from Open to In Review to Matched or Unserviceable. It never expires.

Q: How do I submit a custom request?

A: On Find a Cluster to Reserve click ‘Submit a custom request’ in the banner below the filters, or ‘Submit Custom Request’ when no results match. Your filters are carried over as hints; GPU Model and Geographical Region are required.

Q: Can I cancel a custom request?

A: Yes, while it is Open or In Review. Matched, Unserviceable and Cancelled are final.

Customer — request

Q: How do I create a reservation request?

A: Select a cluster card (or several with the multi-NCP toggle), click Request a Cluster, fill Customer Details, Cluster Details and GPU Reservation Period, then click Submit reservation request. You land on the request detail page.

Q: How do I send a request to multiple NCPs?

A: Turn on ‘Send request to multiple NCP’s’ (requires the send_request_multiple_provider permission), tick up to 3 cluster cards and click Request a Cluster. They are treated as alternatives; you will be matched with one.

Q: What is the maximum number of NCPs in one request?

A: The UI allows 3 by default (NEXT_PUBLIC_MAX_MULTI_NCP_SELECTION_COUNT). The backend cap is 10 (MULTI_PROVIDER_MAX_GPU_OFFERINGS_PER_REQUEST).

Q: Which fields are required in the reservation form?

A: Organization Name, GPU count (at least 1), preferred start date (after today), and either Duration in months (at least 1) or Tenure. Terms are pre-accepted on submit. Organization URL, Contact Email (max 250 chars) and Purpose (max 1,000 chars) are optional.

Q: Can I request more GPUs than a cluster shows?

A: Yes. A yellow warning appears but you can still submit and negotiate with the NCP.

Customer — search

Q: How do I find available GPU clusters?

A: Go to Request a Cluster, which opens Find a Cluster to Reserve. Select a Platform first, then filter by Geographical Region, Country, No. of GPUs, Tenure and Earliest Start Date. Results appear on a world map and as capacity cards.

Q: Why are Tenure and Earliest Start Date disabled?

A: They unlock only after you select a Platform (GPU model).

Q: What do Current and Future mean on a capacity card?

A: Current means available capacity (starting within 90 days). Future means capacity that starts later and was approved by a Platform Admin.

Q: What does the amber note on a cluster card mean?

A: The cluster already has pending requests from other customers.

Q: What tenure options are available?

A: 0-1, 1-2, 2-3, 3-4, 4-5, 5-6 and 6+ years. Availability checks use the upper bound of the band.

Customer — tracking

Q: Where can I see my requests?

A: My requests (/customer/reservations). It lists each request with status, platform, region, GPUs and dates. Multi-NCP requests are grouped with a child row per NCP. Click a row for details.

Q: How do I see why an NCP rejected my request?

A: In My requests, hover the info icon next to the Rejected status, or open the request detail page where rejection remarks appear in the Reservation Summary.

Q: How do I cancel a reservation request?

A: Open the request detail page, click Cancel, and click Confirm in the modal (optionally give a reason). Cancel is available while Awaiting NCP Acceptance, awaiting approval, In Process or Expired. NCPs are notified.

Q: Can I accept an NCP counteroffer from the request page?

A: No. The customer detail page is for viewing and cancelling only. You agree terms with the NCP after the introduction email and the NCP confirms the contract.

Q: What happens after I submit a request?

A: You get a ‘request received’ email. Each NCP gets a magic link and has 7 days to accept. When one accepts, the request is In Process and you get an introduction email; there are 45 days to contract. When an NCP confirms the contract the request is Contracted and the others are cancelled.

Developer — api

Q: Where are the API docs?

A: Locally at http://localhost:8082/docs (Swagger). A snapshot is in backend/openapi.json, which omits Access Request and internal routes.

Q: Which endpoint creates a reservation request?

A: POST /api/v1/reservation-request with scope create_reservation_request.

Q: Which endpoints do NCPs use to act on offerings?

A: POST /api/v1/provider/offering/{id}/accept, /reject and /contract, all requiring approver_manage_reservation_request.

Developer — architecture

Q: What is the tech stack?

A: Frontend: Next.js 15, React 19, Ant Design 6, Tailwind 4, NextAuth 5. Backend: Python 3.13 FastAPI with DDD/hexagonal architecture, SQLAlchemy, PostgreSQL, Alembic. Workflows: Temporal. Identity: Keycloak with Starfleet. Analytics: Superset. Cloud: GCP GKE Autopilot, Cloud SQL, Memorystore Redis, Secret Manager, via Terraform, Helm and GitLab CI.

Q: How is the backend structured?

A: Hexagonal architecture. backend/core holds the domain (bounded contexts like gpu_offering, reservation_request, organization, user) with ports and services. backend/infrastructure/driving/http is the FastAPI adapter; backend/infrastructure/driven holds database, email, Keycloak, Temporal and Salesforce adapters.

Q: What does the workflow worker do versus the backend?

A: The backend owns domain changes. The Temporal worker owns timers, schedules, retries and fan-out, and its activities call backend System APIs with a service account (scopes notification:send, capacity:sync, reservation-request:write).

Developer — delivery

Q: How are releases promoted across environments?

A: Build once, promote digests: main publishes images, Dev auto-deploys and snapshots, QA promotes exact digests into a computematch-release package, Staging deploys that package without rebuilding. Production promotion is not yet enabled.

Developer — git

Q: What is the branch naming convention?

A: NCPMS-/, for example NCPMS-565/repository-foundation. One Jira issue per branch, branched from an updated main.

Q: What is the commit message format?

A: (): NCPMS- , for example fix(backend): NCPMS-855 widen hosting provider name columns. Types: feat, fix, docs, test, refactor, perf, build, ci, chore, revert; add ! after the scope for breaking changes.

Q: What labels must a merge request have?

A: Exactly one type:: label, exactly one risk:: label (low, medium, high) and at least one component: label. Add review:security, review:database, review:platform or breaking-change when relevant.

Developer — infrastructure

Q: Where does Compute Match run in the cloud?

A: Private GKE Autopilot clusters on Google Cloud with Cloud SQL PostgreSQL, Memorystore Redis, Secret Manager, a regional Gateway and Cloud Armor, plus Temporal Cloud.

Developer — keycloak

Q: What custom Keycloak extensions exist?

A: accept-reservation-token (magic-link action token), audit-event-listener (login/logout audit to backend), default-role-restriction (strip default roles from admin-created users) and multi-org-policy (only account managers in multiple organizations).

Developer — local

Q: How do I run Compute Match locally?

A: Install Git, Make and Docker Desktop, connect to the NVIDIA VPN, run make onboard from the repo root (first build 10-20 minutes), then make verify. Open http://localhost:3000 and sign in with a demo account from local-setup/.onboarding/accounts.md.

Q: What are the local demo accounts?

A: Customer developer@customer-a.example.org, provider seed@singtel.example.org, platform business admin business-admin@platform.example.org, operations ops@platform.example.org. Passwords are in local-setup/.onboarding/accounts.md.

Q: How do I develop only the backend on my laptop?

A: Run make deps MODE=backend once, then make backend. The frontend and worker stay in Docker. Return everything to Docker with make start.

Q: What ports do the local services use?

A: App 3000, backend 8082 (/docs), Keycloak 8080, Temporal UI 7600, Temporal gRPC 7233, PostgreSQL 5432, Redis 6379, Superset 8088, SigNoz 3301 (OTLP 4317/4318).

Q: My local setup is broken. How do I reset it?

A: Run make clean && make onboard && make verify. First check make status, make doctor and make logs SERVICE=backend.

Q: I signed up locally but see no features.

A: New sign-ups may need a business role. In Keycloak (http://localhost:8080/admin), choose the customer realm, find the user, assign a role under Role mapping, then sign out and back in.

Developer — observability

Q: What logging format do services use?

A: Log Record v2: JSON Lines on stdout with log_schema_version 2, severity, event_name, resource, source, optional trace context, attributes and exception. No PII or secrets in attributes.

Developer — salesforce

Q: How does Compute Match integrate with Salesforce?

A: One-way sync to the Salesforce Offtake object using a transactional outbox (salesforce_create_sync) drained by SfdcCreateBatchWorkflow: nightly create at 05:00 UTC and status updates every 2 minutes. One Offtake per NCP offering. Partner email domains are also read from Salesforce twice a day.

Developer — testing

Q: How are end-to-end tests written?

A: With Playwright and Cucumber (Gherkin) in e2e_playwright/: features for Authentication, Customer, Provider and Admin workflows, page objects in src/PageObjects and steps in src/Specs. Choose an environment with TEST_ENV.

Developer — workflows

Q: Which Temporal workflows exist?

A: ReservationRequestWorkflow, OfferingLifecycleWorkflow, CsvImportWorkflow, ProviderOnboardingWorkflow, InProcessAlertWorkflow, AuditRetentionWorkflow, SfdcCreateBatchWorkflow (create and status modes) and SfdcPartnerDomainSyncWorkflow.

Q: What signals does OfferingLifecycleWorkflow accept?

A: provider_accepted, acceptance_expiry_extended, contract_expiry_extended, offering_contracted, offering_rejected and offering_cancelled.

Ncp — dashboard

Q: What does the NCP Dashboard show?

A: A read-only overview: a welcome header with how many requests await action, Awaiting Your Action, Available Capacities with your latest upload status, and quick links View all requests, Manage Reservations, View all uploads and Upload Capacity.

Ncp — magic-link

Q: What is a magic link?

A: A secure one-click email link sent to NCP users when a request is routed to their organization. It logs them in through a Keycloak action token and opens Manage reservation so they can accept.

Q: Who receives magic-link emails?

A: Users in the NCP organization who have the provider_business_user role and belong to the Keycloak group offering-validator. Basic users do not.

Q: My magic link has expired. What do I do?

A: Magic links are valid for 1 hour. Request a resend (allowed for 7 days after routing) or simply log in and open Manage reservation.

Q: Why does the magic link say a different user is logged in?

A: The accept-reservation-token Keycloak extension blocks a link if another user is already signed in. Log out and click the link again.

Ncp — manage

Q: How do I accept a reservation request as an NCP?

A: Open Manage reservation (/provider?view=split), choose the Awaiting NCP Acceptance tab, find the card and click Accept. The offering becomes In Process and a 45-day contract window starts.

Q: How do I reject a request?

A: Click Reject on the card in Manage reservation and confirm. Remarks are required and are shown to the customer.

Q: How do I make a counteroffer?

A: Click Counteroffer on the card. The Modify Details panel shows the original request on the left and editable Proposed Start Date, Proposed End Date, Number of GPUs, Platform and

Geographical Region on the right. Click Confirm (or Reset / Cancel).

Q: How do I finalize a contract?

A: For an In Process card, click Confirm Contract. The offering and request become Contracted, capacity is allocated, and other NCPs are cancelled as superseded.

Q: Why did Accept or Confirm Contract fail with capacity not available?

A: Your remaining capacity for that offering, region and capacity type no longer covers the requested GPUs and period (availability is checked at accept and contract time).

Q: When does the capacity chart appear in Manage reservation?

A: After you select Platform, Geographical Region and Country in the left panel filters.

Q: What is the difference between Manage reservation and Reservation requests?

A: Manage reservation is the split view with action cards and a capacity chart. Reservation requests is a sortable table for browsing; to act, choose filters and click Manage Reservation.

Q: Can I reorder reservation cards?

A: Yes, drag cards to set priority, or use Space/Enter to grab and arrow keys to move.

Ncp — roles

Q: What is the difference between provider_business_user and provider_basic_user?

A: provider_business_user can manage inventory and act on reservations, and (with the offering-validator group) receives magic links. provider_basic_user is read-only, never receives magic links and only gets the table view.

Q: Why is the customer name shown as Confidential?

A: Customer organization names can be hidden from NCPs and shown blurred as Confidential until engagement, to protect customer privacy.

Ncp — sign-in

Q: Why does it say ‘Your access is being reviewed’?

A: Your email domain belongs to an NVIDIA partner but your NCP organization is not onboarded yet. An access request was created and a Platform Admin must approve it. When approved, the page shows ‘Your access has been approved’ while your session refreshes.

Q: How do NPN members sign in?

A: NPN (NVIDIA Partner Network) members can reach Compute Match directly from the NPN portal. They still enter their work email and sign in through Starfleet. If their NCP is not onboarded, an access request with source NPN_PORTAL is created for admin approval.

Ncp — upload

Q: How do I upload GPU capacity?

A: Go to Upload Capacities, click Download Template, fill it, then drag and drop or select the CSV. After Upload Successful, watch progress in Upload History.

Q: What columns does the capacity CSV need?

A: Required: GPU Platform Type, Number of GPUs Available, Live Token Date, Country, City. Optional: Region, end_datetime. NCP Name, RA/RD, Exemplar Cluster, aaS Capability and State are accepted but ignored.

Q: What are the CSV size limits?

A: UTF-8, at most 10 MB, 10,000 data rows and 500 characters per cell. (The in-app manual still says 1,000 rows, but the backend enforces 10,000.)

Q: Why is my upload Pending Review?

A: It contains future capacity — rows whose Live Token Date is more than 90 days after upload. A Platform Admin must accept or reject it. Current-capacity rows already went live.

Q: What happens to invalid rows in my CSV?

A: They are skipped and listed as errors (see all details in Upload History). The job only fails if no row is valid.

Q: Do I need to put my hosting provider name in the CSV?

A: No. The hosting provider name comes from your NCP organization.

Q: What are the CSV import job statuses?

A: Backend: QUEUED, VALIDATING, VALIDATION_FAILED, IMPORTING, IMPORT_FAILED, PENDING_REVIEW, APPROVED, REJECTED, FUTURE_CAPACITY_REJECTED, COMPLETED. UI labels: Processing, Pending Review, Completed, Rejected, Validation Failed.

Q: What is the difference between current and future capacity?

A: Current (AVAILABLE) capacity starts within 90 days of upload and is published right after validation. Future (FUTURE) capacity starts more than 90 days out and needs Platform Admin review before customers see it.